

Digital Clock: Module Construction and Project Details.

Alasdair Nicol. S.N:1759586

1/5/2026

Contents

0.1	Project Setup	2
0.1.1	Backup	2
0.2	Modules	2
0.2.1	Module 1: Seven Segment	3
0.2.2	Module 2: Up Down Counter	3
0.2.3	Module 3: Binary to BCD	4
0.2.4	Module 4: HMS Counter	5
0.3	Report Qestions	6
0.3.1	Report 1	6
0.3.2	Report 4	6

0.1 Project Setup

This section covers key information about the project settings, and backup settings for the project.

0.1.1 Backup

Document was backed up both locally and in an online repository using the git command-line tool. A backup was made at the end of each module design cycle and before closing the project workspace.

0.2 Modules

This section covers the function and testing of each module in the project.

0.2.1 Module 1: Seven Segment

Waveform

The waveform in figure 1 demonstrates the correct 7f output (1111111) for the active low display when the blank signal is high. Otherwise, the output displays the correct bit encoding for the given input value.

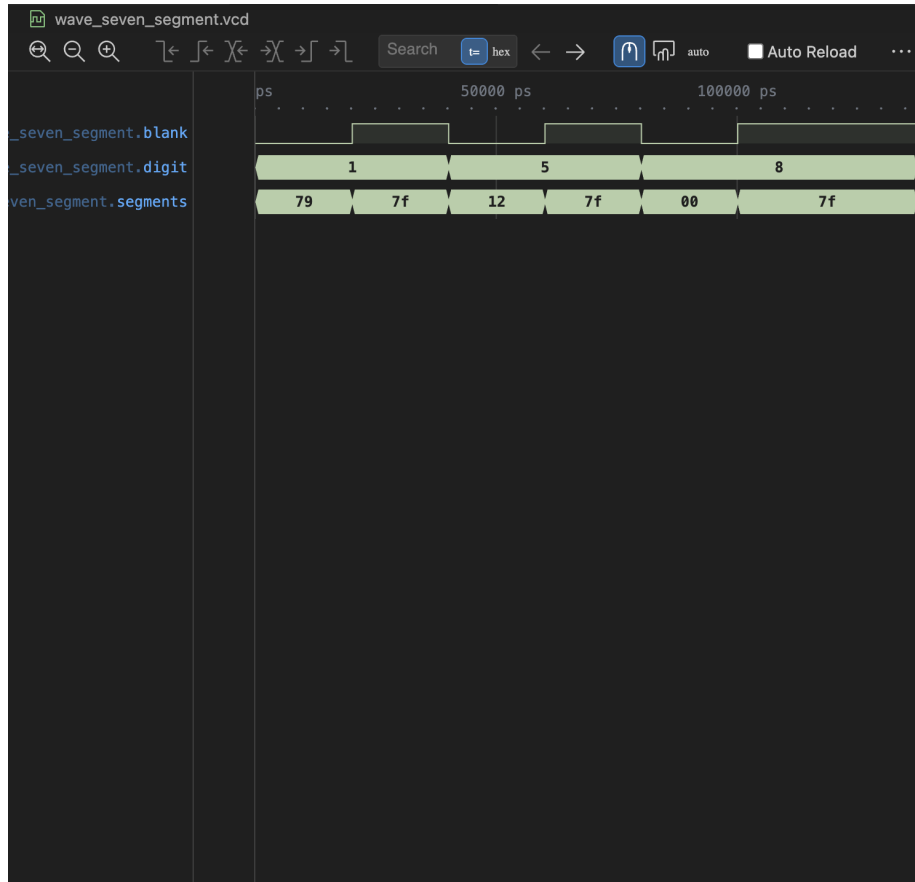


Figure 1: Waveform generated for the seven segment module.

0.2.2 Module 2: Up Down Counter

Waveform

When the enable signal is low, the counter correctly holds onto the current value of 0 (as seen in figure 2), before counting up in increments of one when both the enable pin and up pin are high. Furthermore, the timer correctly counts backwards with a unit of one at the next clock signal after the up signal goes low.

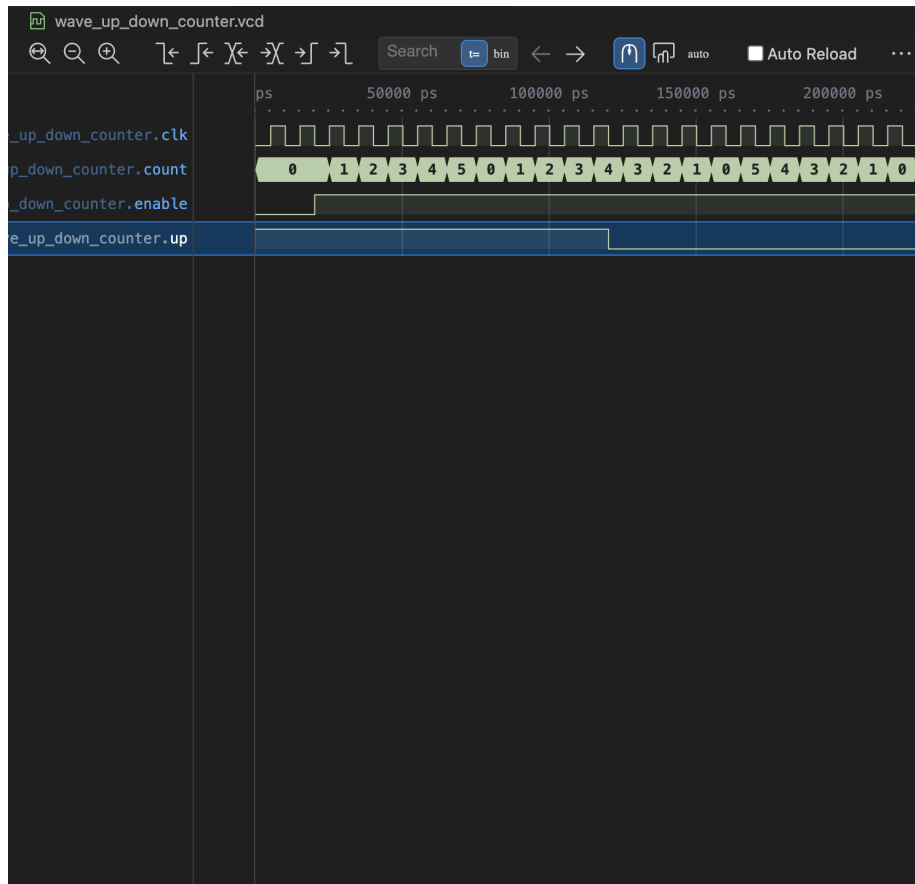


Figure 2: Waveform generated for the up down counter.

0.2.3 Module 3: Binary to BCD

Waveform

As can be seen with the bin signal of 63 in figure 3, the circuit correctly breaks down the hexadecimal 63(decimal 99) into the ones and tens components of 9 and 9.

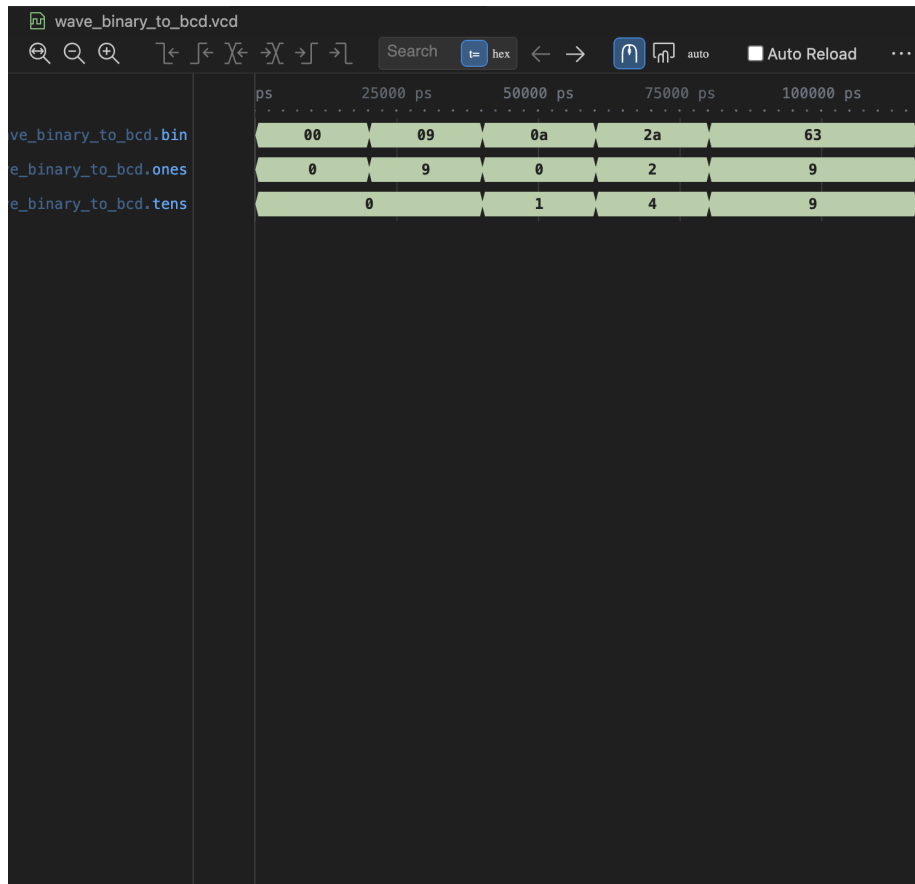


Figure 3: Waveform generated for the binary to bcd converter.

0.2.4 Module 4: HMS Counter

Waveform

The seconds signal of figure /reffig:hms successfully iterates to 4 on the rising edge of the rising edge of the clock, where it then switches to 0 and triggers an iteration in the minutes signal. This is the expected defined bahvaiour given the bit width and max count values used to test the circuit. The enable signal is also seen to hold the seconds signal at 0 from time 0 to the high enable signal.

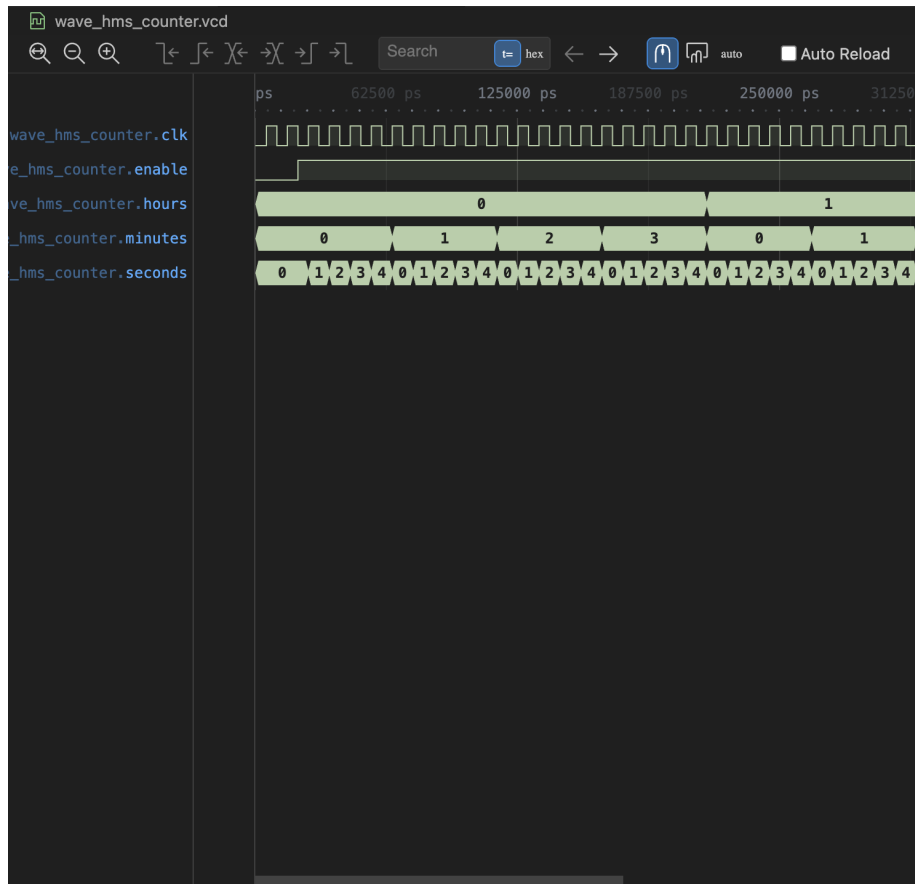


Figure 4: Waveform generated for the binary to bcd converter.

0.3 Report Questions

0.3.1 Report 1

Blank is a 1 bit wide logic signal, while ACTIVE LOW is defined as an integer, which is a multi-bit data type (32 by default) which will throw a warning if truncated.

0.3.2 Report 4

BCD is a method of encoding decimal bits in binary that reserves a number of bits (usually 4 for 0-9) for each decimal digit, before concatenating all of the digits into a single register. Storing the digits individually provides greater accuracy at high counts, and helps to avoid rounding errors making it a common tool in precision counters.